

ECON 351 · Midterm Simulation

Practice Test 2

24 Questions | 24 points (1 pt each)

FORMAT (matches your teacher's breakdown):

PART A · Chapter 4 (12 questions)

- 2 Consumer Problems
- 2 Labor-Leisure
- 2 Intertemporal Consumption
- 3 Elasticity / Types of Goods
- 1 Budget Line
- 2 Assumptions / Preferences

PART B · Chapter 9 (12 questions)

- 5 Insurance Problems
- 4 Risk Attitudes
- 2 Certainty Equivalent & Risk Premium
- 1 Diversification

Problems from HW, sample questions, and mock midterm (screenshots).

Use Formula_Sheet.pdf for reference.

PART A

CHAPTER 4

12 questions



Question 26:

There are two goods, apples and bananas. The price of apples is $P_A = \$2$, and the price of bananas is $P_B = \$3$. A consumer has \$120 to spend, and his utility function is $U(A, B) = 2A^2B^3$.

a) Write the expression that represents the consumer's budget constraint. Graph the budget constraint, and label it "BC" (apples should be on the horizontal axis). What is the slope of the budget constraint?

b) Find the consumption basket that maximizes the consumer's utility. Plot this point on the graph, and label it as "Point X". Draw the indifference curve that goes through Point X, and label it "IC" (the indifference curve can be an approximation, it does not have to be exact). What is the slope of the indifference curve at Point X?

Question 27:

Adam derives utility from consuming apples and bananas. His marginal utility from consuming apples is

$$MU_A = 5A^4B^3,$$

while his marginal utility from consuming bananas is

$$MU_B = 3A^5B^2,$$

where A is the quantity of apples and B is the quantity of bananas. We know that the price of apples is $P_A = 6$, the price of bananas is $P_B = 2$ and Adam's income is $I=240$. Given this information, when Adam optimally chooses his consumption, he buys how many units of apples?

(A) A=65

(B) A=45

(C) A=25

(D) A=35

(E) A=55

Question 28:

Monica only consumes goods A and B. Suppose that her marginal utility from consuming good A is equal to $1/Q_A$, and her marginal utility from consuming good B is $1/Q_B$. If the price of A is \$0.50, the price of B is \$4.00, and the Monica's income is \$120.00, how much of good A will she purchase?

(A) 0

(B) 12

(C) 24

(D) 48

(E) 120

[Q021]

Maria: 16 hrs/day, $w=\$20$, $V=\$80$, $U=c^{0.5} l^{0.5}$. Find I^* , L^* , c^* .
(Course-style)

[Q022]

Same setup, $w=\$30$. Find I^* , L^* , c^* . More or less work?
(Course-style)

Question 5 of 24 | Ch. 4

Intertemporal Consumption

[Q023]

Contract 1: \$10M today + \$90M future. Contract 2: \$42M + \$54M. $r=0.20$. Cheaper PV?

HW1 Ex 5.1

Question 24:**[Show your work]**

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 60$ in the first period and $I_2 = 125$ in the second period. The consumer can borrow or save money at the interest rate of 25%, that is, $r=0.25$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

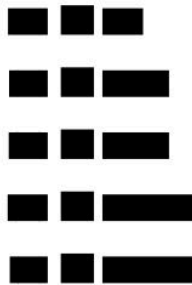
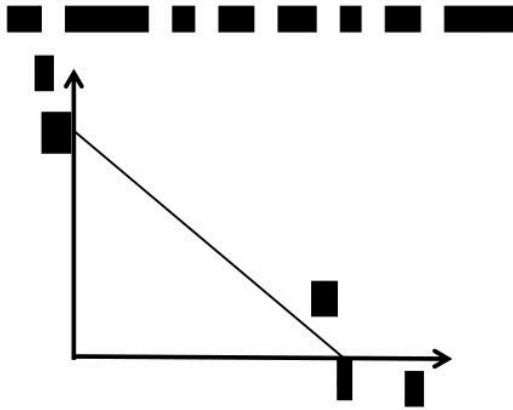
The consumer's inter-temporal budget constraint is

- (A) $q_1 + 0.8q_2 = 160$
- (B) $q_1 + q_2 = 185$
- (C) $q_1 + 1.25q_2 = 216.25$
- (D) $q_1 + 0.75q_2 = 160$
- (E) $q_1 + 0.25q_2 = 200$

Question 25:**[Show your work]**

Consider all the information from Question 24. In the first period, the consumer will optimally purchase

- (A) $q_1 = 0$
- (B) $q_1 = 100$
- (C) $q_1 = 60$
- (D) $q_1 = 80$
- (E) $q_1 = 120$

**Question 34:**

The price of good A goes up. As a result, the demand for good B shifts to the left. From this we can infer that:

- (A) good A is a normal good.
- (B) good B is an inferior good.
- (C) goods A and B are substitutes.
- (D) goods A and B are complements.
- (E) none of the above

Question 35:

The price of good A goes down. As a result, the demand for good B shifts to the left. From this we can infer that:

- (A) good A is a normal good.
- (B) good B is an inferior good.
- (C) goods A and B are substitutes.
- (D) goods A and B are complements.
- (E) none of the above

Question 36:

The change in the price of one good has no effect on the quantity demanded of another good. These goods are:

- (A) complements.
- (B) substitutes.
- (C) both inferior.
- (D) both elastic.
- (E) none of the above.

Question 37:

If goods X and Y are substitutes, then their cross-price elasticity is

- (A) positive.
- (B) negative.
- (C) zero.
- (D) +1 if they are perfect substitutes.
- (E) -1 if they are perfect substitutes.

Question 38:

A consulting company researched the market for apples in California, and concluded the following: apples and bananas are complements, and apples are an inferior good.

Define the following:

Q_A : quantity of apples demanded

P_A : price of Apples

P_B : price of bananas

I : income

From the options below, mark the only demand function that can possibly represent the demand for apples in California.

(A) $Q_A = 100 - 3P_A + 3I - 5P_B$

(B) $Q_A = 200 - P_A + 2I + 4P_B$

(C) $Q_A = 110 - 2P_A - 4I + 2P_B$

(D) $Q_A = 250 - 4P_A - 5I - 3P_B$

(E) $Q_A = 150 + P_A + I + P_B$

Question 39:

The demand for apples (Q_A) is given by the function

$$Q_A = 784 - 2P_A - 3I - 4P_O$$

where P_A is the price of apples, I is the income level, and P_O is the price of oranges.

Suppose $P_A = 2$, $I = 100$, and $P_O = 20$.

a) Compute the price elasticity of demand for apples.

b) Compute the income elasticity of demand for apples. Are apples a normal good or inferior good?

c) Compute the cross-price elasticity of demand for apples. Are apples and oranges substitutes or complements?

Question 40:

The income elasticity of demand for good X is -2. Due to a financial crisis, the income of consumers declined by 2%. How did the financial crisis affect the quantity demanded for X?

(A) The quantity demanded increased 2%

(B) The quantity demanded decreased 2%

(C) The quantity demanded increased 4%

(D) The quantity demanded decreased 4%

(E) The quantity demanded did not change

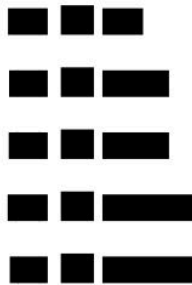
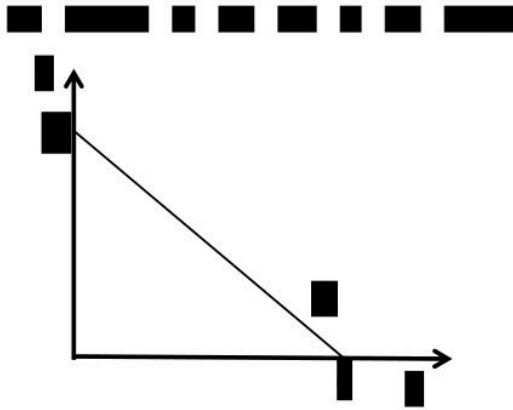
Question 41:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 50%, that is, $r = 0.5$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

What is the optimal choice of q_1 and q_2 ? Is this consumer saving or borrowing? How much?

Question 42:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 5%, that is, $r = 0.05$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

**Question 34:**

The price of good A goes up. As a result, the demand for good B shifts to the left. From this we can infer that:

- (A) good A is a normal good.
- (B) good B is an inferior good.
- (C) goods A and B are substitutes.
- (D) goods A and B are complements.
- (E) none of the above

Question 35:

The price of good A goes down. As a result, the demand for good B shifts to the left. From this we can infer that:

- (A) good A is a normal good.
- (B) good B is an inferior good.
- (C) goods A and B are substitutes.
- (D) goods A and B are complements.
- (E) none of the above

Question 36:

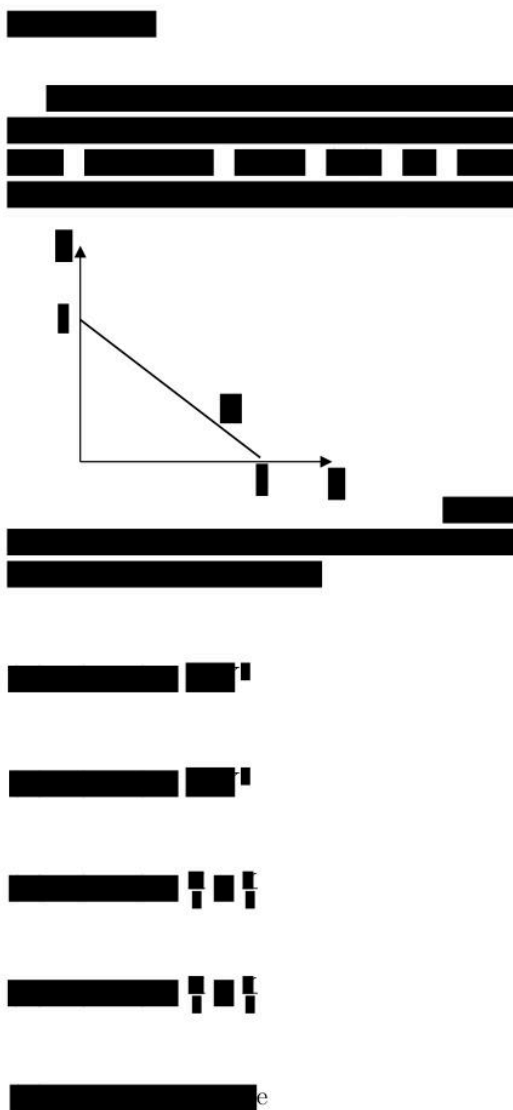
The change in the price of one good has no effect on the quantity demanded of another good. These goods are:

- (A) complements.
- (B) substitutes.
- (C) both inferior.
- (D) both elastic.
- (E) none of the above.

Question 37:

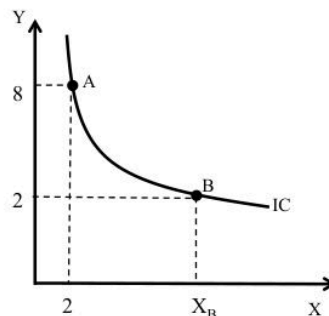
If goods X and Y are substitutes, then their cross-price elasticity is

- (A) positive.
- (B) negative.
- (C) zero.
- (D) +1 if they are perfect substitutes.
- (E) -1 if they are perfect substitutes.



Question 9:

Angelique's preferences over goods X and Y can be represented by the following utility function: $u(X, Y) = X^2Y^3$. The graph below represents two baskets, A and B, on the same indifference curve of this consumer.



What do we know about X_B ?

- (A) $X_B = 4$
- (B) $X_B = 3$
- (C) $X_B = 18$
- (D) $X_B = 16$
- (E) $X_B = 8$

Question 10:

Assume that apples are measured on the horizontal axis and oranges on the vertical axis. If the price of apples increases from \$3 to \$6, and the price of oranges increases from \$2 to \$4, then the budget line will:

- (A) become flatter.
- (B) become steeper.
- (C) shift inward (parallel shift).
- (D) shift outward (parallel shift).
- (E) become steeper or flatter depending on the relationship between prices and income.

[Q001]

Jackie: 'Knives Out' vs 'Top Gun Maverick.' She doesn't care which. Violation?

HW1 Ex 2.1

Question 4:

Rafinha has the following utility function for goods X and Y: $u(X, Y) = 6X^{0.5}Y^{0.5}$. When Rafinha consumes $X=9$ and $Y=81$, with X on the horizontal axis, what is the absolute value of the slope of his indifference curve?

- (A) 3.
- (B) 6.
- (C) 9.
- (D) 12.
- (E) 15.

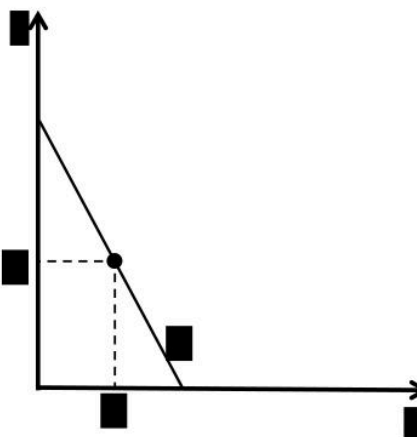
(D) all of the above.

(E) A and C only.

Question 5:

By observing the slope of an indifference curve, we can infer

- (A) that preferences are complete.
- (B) the marginal utility of the good on the vertical axis.
- (C) the marginal rate of substitution of one good for another good.
- (D) the ratio of market prices.
- (E) that preferences are transitive.

**Question 6:**

The magnitude of the slope of an indifference curve is:

- (A) called the marginal rate of substitution.
- (B) equal to the ratio of the total utility of the goods.
- (C) always equal to the ratio of the prices of the goods.

PART B

CHAPTER 9

12 questions

[Q049]

Tom insurance: risk premium = max premium - fair premium.

HW2 Ex 3c

Question 16:

Using the information from the previous question, how much is the fair insurance premium?

- (A) 174,000
- (B) 72,000
- (C) 130,500
- (D) 43,500
- (E) 168,100

Question 17:

Using the information from Question 15, find Tom's associated risk premium:

- (A) 1,200
- (B) 2,100
- (C) 1,550
- (D) 2,000
- (E) 1,875

Question 18:

Sherman has the following utility from wealth:

$$u(x) = \sqrt{x}.$$

Wealth is the sum of cash and the value of his car. He currently has a Ferrari worth \$230,900 and \$50,000 in cash (risk free). With probability 25%, a paparazzi will cause \$158,400 in damages to the car (in this case, the car value becomes \$72,500). With probability 75% nothing happens, and the car keeps its original value. AAA offers to fully insure the car for a price P . What is the maximum insurance premium P that Sherman is willing to pay? What is the associated risk premium?

Question 19:

Joe's utility from income is given by

$$u(I) = 8\sqrt{I}.$$

He has a house in LA County. With probability $1/4$, the rain will cause a mudslide and damage his house, decreasing its value to \$250,000. With probability $3/4$, no mudslide occurs and the house retains its original \$1,000,000 value. AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Joe is willing to pay? What is the associated risk premium?

Question 20:

Marcelo has the following utility from income: $u(I) = \sqrt{I}$. He has a \$10,000 car. With probability 25% he will crash his car, and will have to pay \$6,348 to the repair shop to fix it. With probability 75% no crash occurs, and he does not have to repair the car. Progressive Insurance offers to partially insure the car for a price $P=124$. That is, Marcelo pays insurance premium $P=124$, and in case of a crash, Progressive will only pay for half of the repair cost (in case of a crash, Progressive will only pay \$3,174 to Marcelo). If Marcelo buys this partial insurance plan, his expected utility will be:

- (A) 94
- (B) 98
- (C) 96
- (D) 95
- (E) 97

Exercise 3 - Insurance

Tom has the following utility from income:

$$u(I) = \sqrt{x}$$

He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house. In this case, the house value becomes \$672,400. AAA offers to fully insure the house for a price P .

- a) What is the maximum insurance premium P that Tom is willing to pay?
- b) How much is the expected loss? (also known as the fair insurance premium)
- c) How much is the risk premium? (difference between the max.premium the consumer is willing to pay and the expected loss)

Question 27:

[Show your work] A consumer has \$1,000 in cash. With probability 90% nothing happens (he does not get sick) and he is able to keep all his money. With probability 10% he gets sick and must go to the doctor. The cost of going to the doctor is \$500.

An insurance company offers the following health insurance plan. The consumer must pay upfront a premium $p = \$100$. If the consumer gets sick, the insurance company pays the doctor, but the consumer must pay a \$275 copay.

The consumer's utility from income is given by $u(I) = \sqrt{I}$. Compute the expected utility of the consumer if he purchases this insurance.

- (A) 29.5
- (B) 30.7
- (C) 28.6
- (D) 30
- (E) 29

Question 25:

[Show your work] A consumer has the following utility from income: $u(I) = \sqrt{I}$. He bought a car for \$10,000. With probability 75%, a crash will cause \$3,600 in damages to the car (in this case, the car's value becomes \$6,400). AAA offers to fully insure the car for a price P. What is the maximum insurance premium P that the consumer is willing to pay?

- (A) 2,725
- (B) 2,850
- (C) 2,750
- (D) 2,825
- (E) 2,775

Question 26:

[Show your work] Using the information from the previous question, compute the associated risk premium.

- (A) 75
- (B) 125
- (C) 50
- (D) 150
- (E) 25

- a) Compute the expected value of each investment option.
- b) Compute the expected utility of each investment option. Find the best (and worse) investment option for Jim.

Question 3:

Pedro Carroll has the following utility from income: $u(I) = \sqrt{I}$. Currently, he has an income $I = \$169$ in cash (no risk). He is deciding whether or not to bet all his money on the Los Angeles Seahawks to win the championship game. If he bets all his money and the Seahawks win, then he ends up with an income $I = \$400$. If he bets all his money and the Seahawks lose, then he ends up with nothing, $I = \$0$. If he does not bet, he keeps his \$169 for sure.

The Los Angeles Seahawks will win the championship game with probability P , and will lose the game with probability $1-P$.

Compute the minimum winning probability P such that Pedro would be willing to bet all his money on the Seahawks.

- (A) $P = 50\%$
- (B) $P = 65\%$
- (C) $P = 70\%$
- (D) $P = 55\%$
- (E) $P = 60\%$

Question 4:

Cam has the following utility from income: $u(I) = \sqrt{I}$. Currently he has an income $I = \$36$ in cash (no risk). He is deciding whether or not to bet all his money on the Los Angeles Panthers to win the championship game. If he bets all his money and

the Panthers win, then he ends up with an income I_{Win} . If he bets all his money and the Panthers lose, then he ends up with nothing, $I_{Lose} = \$0$. If he does not bet, he keeps his \$36 for sure.

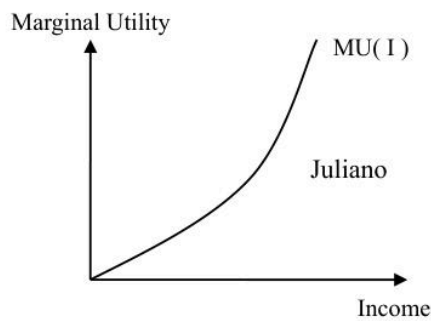
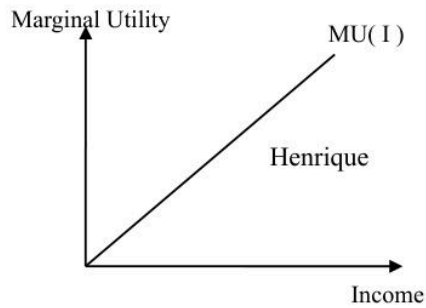
The Los Angeles Panthers will win the championship game with probability 75%, and will lose the game with probability 25%.

Compute the minimum winning prize I_{Win} such that Cam would be willing to bet all his money on the Panthers.

- (A) $I_{Win} = \$36$
- (B) $I_{Win} = \$48$
- (C) $I_{Win} = \$56$
- (D) $I_{Win} = \$64$
- (E) $I_{Win} = \$72$

**Question 6:**

Anna's utility of income function is $U(I) = I$, while Elsa's utility of income function is $U(I) = I^{1.5}$. Therefore, Anna is _____ and Elsa is _____.



Using the information from the graphs, Henrique is _____ and Juliano is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Question 10:

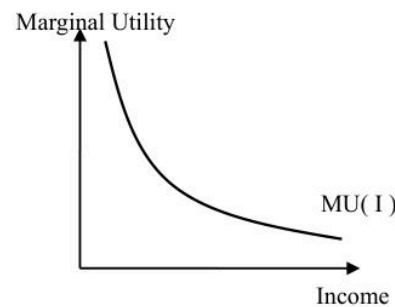
Mike has a constant marginal utility of income. We know that Mike's income I is greater than \$10, but we do not know his exact income. Adam offers Mike the following bet: Adam will throw a fair coin (that is, heads and tails have equal probability). If it is heads, Mike pays Adam \$10. If it is tails, Adam pays Mike \$10. Mike will

- (A) reject the bet only if I is high enough

- (B) reject the bet, independently of I
- (C) accept the bet only if I is high enough
- (D) accept the bet, independently of I
- (E) always be indifferent between accepting and rejecting the bet

Question 11:

The following graph represents Mike's marginal utility of income:



We know that Mike's income I is greater than \$10, but we do not know his exact income. Adam offers Mike the following bet: Adam will throw a fair coin (that is, heads and tails have equal probability). If it is heads, Mike pays Adam \$10. If it is tails, Adam pays Mike \$10. Mike will

- (A) reject the bet only if I is high enough
- (B) reject the bet, independently of I
- (C) accept the bet only if I is high enough
- (D) accept the bet, independently of I
- (E) always be indifferent between accepting and rejecting the bet

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

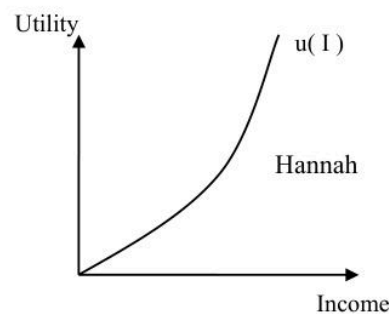
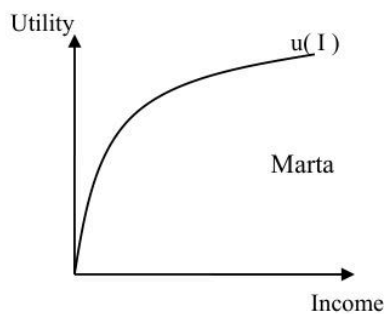
Question 7:

Anna's marginal utility of income function is $MU(I) = I$, while Elsa's marginal utility of income function is $MU(I) = I^{1.5}$. Therefore, Anna is _____ and Elsa is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Question 8:

The following graphs represent the utility function of consumers Marta and Hannah:



Using the information from the graphs,

- (A) Both Marta and Hannah are risk loving
- (B) Both Marta and Hannah are risk averse
- (C) Marta is risk averse; Hannah is risk loving
- (D) Marta is risk loving; Hannah is risk averse
- (E) Not enough information from the graphs

Question 9:

The following graphs represent the marginal utility of income functions of consumers Henrique and Juliano.

Question 28:

Firmino is a big fan of a soccer team called The Reds. Everyone knows that there is a 75% chance that the The Reds will win the next tournament, and a 25% chance that they will lose. Firmino offers the following bet to each one of his friends: if The Reds win, then the friend must pay \$1 to Firmino. If The Reds lose, then Firmino will pay \$3 to the friend.

The friends Mohamed and Sadio (who are very rich) are each considering whether to accept the bet. Mohamed's utility of income function is

$$U(I) = I^{0.7}$$

while Sadio's utility of income function is

$$U(I) = I^{2.3}$$

Therefore,

- (A) Both Mohamed and Sadio will reject the bet
- (B) Both Mohamed and Sadio will accept the bet
- (C) Mohamed will accept and Sadio will reject the bet
- (D) Mohamed will reject and Sadio will accept the bet
- (E) Both Mohamed and Sadio are indifferent between accepting and rejecting the bet

Question 29:

Assets A and B have positively correlated returns, while assets C and D have negatively correlated returns. John is an investor who has chosen to invest half of his money on asset A and half on asset B. Marc is an investor who has chosen to invest half of his money on asset C and half on asset D. Therefore, which

of the following options is most likely to be true when we compare John and Marc?

- (A) John's investment has a lower variance and John is more risk averse than Marc
- (B) John's investment has a higher variance and John is more risk averse than Marc
- (C) John's investment has a lower variance and John is less risk averse than Marc
- (D) John's investment has a higher variance and John is less risk averse than Marc
- (E) Both investment portfolios have the same variance and the investors are risk neutral.

Question 30:

Mirabel's marginal utility of income function is $MU(I) = 9$, while Bruno's marginal utility of income function is $MU(I) = \frac{3}{I}$. Therefore, Mirabel is _____ and Bruno is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk neutral, risk averse.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Question 12:

A consumer owns shares of a company. With probability 80% the value of the investment will go up to $x = \$144$, and with probability 20% the value of the investment will go down to $x = \$81$. The consumer's utility from income is given by the function

$$u(x) = \sqrt{x}.$$

- Compute the consumer's expected utility from holding this investment.
- Compute the expected value of this investment.
- Compute the certainty equivalent of this investment.
- Compute the risk premium associated with this investment.

Question 13:

Cuban is an investor who used all his income to buy a startup company called Shark Aquarium. With probability 50% the value of the company will increase to \$144. With probability 50% the value of the company will go down to \$64. Cuban's utility from income is

$$u(I) = \sqrt{I}.$$

He has no other source of income. Compute the minimum price P such that Cuban would be willing to sell all his shares of Shark Aquarium.

- $P = 94$
- $P = 130$
- $P = 121$
- $P = 104$
- $P = 100$

Question 14:

Using the information from the previous question, compute the associated risk premium.

- 3
- 1
- 7
- 10
- 4

Question 15:

Tom has the following utility from income: $u(I) = \sqrt{I}$. He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house (in this case, the house value becomes \$672,400). AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Tom is willing to pay?

- 23,010
- 45,375
- 74,000
- 32,375
- 15,600

Question 24:

[Show your work] Cuban is a risk-loving investor who used all his income to buy a startup company called Shark Aquarium. With probability 50% the value of the company will increase to \$144. With probability 50% the value of the company will go down to \$64. Cuban's utility from income is

$$u(I) = I^2.$$

He has no other source of income. Compute the minimum price P such that Cuban would be willing to sell all his shares of Shark Aquarium. This P is approximately

- (A) $P = 100$
- (B) $P = 130.1$
- (C) $P = 121.2$
- (D) $P = 111.4$
- (E) $P = 104$

Question 28:

Firmino is a big fan of a soccer team called The Reds. Everyone knows that there is a 75% chance that the The Reds will win the next tournament, and a 25% chance that they will lose. Firmino offers the following bet to each one of his friends: if The Reds win, then the friend must pay \$1 to Firmino. If The Reds lose, then Firmino will pay \$3 to the friend.

The friends Mohamed and Sadio (who are very rich) are each considering whether to accept the bet. Mohamed's utility of income function is

$$U(I) = I^{0.7}$$

while Sadio's utility of income function is

$$U(I) = I^{2.3}$$

Therefore,

- (A) Both Mohamed and Sadio will reject the bet
- (B) Both Mohamed and Sadio will accept the bet
- (C) Mohamed will accept and Sadio will reject the bet
- (D) Mohamed will reject and Sadio will accept the bet
- (E) Both Mohamed and Sadio are indifferent between accepting and rejecting the bet

Question 29:

Assets A and B have positively correlated returns, while assets C and D have negatively correlated returns. John is an investor who has chosen to invest half of his money on asset A and half on asset B. Marc is an investor who has chosen to invest half of his money on asset C and half on asset D. Therefore, which

of the following options is most likely to be true when we compare John and Marc?

- (A) John's investment has a lower variance and John is more risk averse than Marc
- (B) John's investment has a higher variance and John is more risk averse than Marc
- (C) John's investment has a lower variance and John is less risk averse than Marc
- (D) John's investment has a higher variance and John is less risk averse than Marc
- (E) Both investment portfolios have the same variance and the investors are risk neutral.

Question 30:

Mirabel's marginal utility of income function is $MU(I) = 9$, while Bruno's marginal utility of income function is $MU(I) = \frac{3}{I}$. Therefore, Mirabel is _____ and Bruno is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk neutral, risk averse.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.